CIRC HQ STUDENT HANDOUT | GRADE 5

Biomimicry Grabber

Goal: Design a gripping mechanism inspired by how a plant or animal holds, hooks, wraps, or adheres.

35 minutes | 8 steps | Engineering and Design, Library and Information Literacy | Playbook B meeting 19, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Demonstrate the grip and connect one design feature to the biological function it models.

You will use

- Nature reference cards
- Cardboard
- Craft sticks
- String
- Rubber bands
- Foam
- Lightweight test objects
- Scissors

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Choose one nature reference card.
- Place lightweight objects on the test mat.

2 Step 2: Grabber safety — 3 min

- Cut only on the table.
- Keep fingers outside the cutting path.
- Carry scissors closed.
- Test no body or live organism.

3 Step 3: Copy a strategy — 5 min

- Name the nature trick.
- Name its useful function.
- Sketch a safe gripper.

4 Step 4: Build the grabber — 8 min

- Cut only planned cardboard shapes.
- Build the moving connection.
- Add a soft place that can hold an object.

5 Step 5: Test three objects — 7 min

- Grip the first approved object.
- Record the result.
- Repeat for two more objects.

6 Step 6: Improve the grip — 4 min

- Change one nature-inspired part.
- Repeat one failed object test.

7 Step 7: Reset the grabber — 2 min

- Close the scissors.
- Return every test object.

8 Step 8: Exit check — 3 min

- Name the copied strategy.
- Cite one test result.

What to turn in

Your grabber and the animal or plant idea it copies.

No computer? Another way to do it

Draw the grabber and explain the nature idea.

Finished early? Next challenge

Habitat Shift: Adapt the gripper for an object with a different surface, shape, or location.

Stay safe

Use no live organism, body grip test, sharp hook, or adhesive on skin or personal property.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.